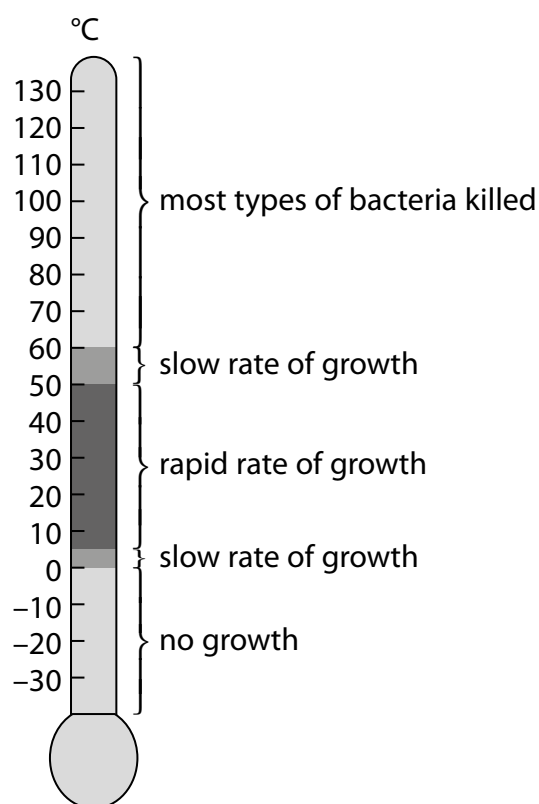


7 Temperature affects the rate of growth of bacteria.

The diagram shows some information about the growth of bacteria in different ranges of temperature.



- (a) Explain why most types of bacteria are killed at temperatures above 60 °C, but bacteria can grow slowly in a temperature range of 50 °C to 60 °C.

(4)

[illegible]

- (b) (i) The growth rate constant will be at its highest in the temperature range of 5 °C to 50 °C.

Calculate the growth rate constant (k) of bacteria that have increased from 5×10^3 cells per cm^3 to 1.3×10^5 cells per cm^3 in 4 hours.

(3)

$$k = \frac{\log_{10} N_t - \log_{10} N_0}{0.301 \times t}$$

Answer.....

- (ii) The formula used to calculate the growth rate constant can only be applied to one phase of bacterial growth.

To which phase of bacterial growth can the formula be applied?

(1)

- ☐ **A** death
- ☐ **B** exponential
- ☐ **C** lag
- ☐ **D** stationary

(c) (i) Explain why some foods are kept in refrigerators at a temperature between 0°C and 5°C .

(3)

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(ii) Explain why there is no growth of bacteria in a freezer at a temperature of -18°C .

(2)

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(Total for Question 7 = 13 marks)